



# Time Series Analysis

## Seminar 8: Long memory and ARFIMA



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

## Acronyms used in this material (1/2)

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ CPI: Consumer Price Index
- ▣ CPIAUCSL: Consumer Price Index for All Urban Consumers (FRED series code)
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPH: Geweke–Porter–Hudak (estimator)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ LW: Local Whittle (estimator)
- ▣ MA: Moving Average
- ▣ ML: Machine Learning
- ▣ OLS: Ordinary Least Squares
- ▣ RV: Realised Volatility
- ▣ SE: Standard Error

## Acronyms used in this material (2/2)

□ **US: United States**

## Today's question and route

- **Question:** does a shock to a series fade at an exponential rate (ARMA) or much more slowly (long memory), and how do we measure the difference?
  - ▶ this seminar comes **before** Lecture 8: the section “Prerequisites for Today” gives every definition the tasks use
- Route
  - ▶ Part A: fractional weights and ARFIMA autocorrelations; R/S and GPH by hand; an ARFIMA forecast; GARCH and FIGARCH weights
  - ▶ Part B: the Nile and its break; Romanian inflation; S&P 500 volatility; BET and EUR/RON
  - ▶ Part C: US inflation and monetary regimes; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

## Exercise map

| Task   | Question                                                          | Type                | Model  |
|--------|-------------------------------------------------------------------|---------------------|--------|
| A1, A2 | weights of $(1 - L)^d$ , ACF of ARFIMA(0, $d$ , 0), ranges of $d$ | Solved,<br>Proposed | A1     |
| A3, A4 | R/S and $H$ by hand; a GPH slope by hand                          | Solved,<br>Proposed | A3     |
| A5, A6 | an ARFIMA forecast; GARCH and FIGARCH weights                     | Solved,<br>Proposed | A5     |
| B1, B2 | the Nile and its break; Romanian inflation                        | Solved,<br>Proposed | B1     |
| B3, B4 | S&P 500 volatility; BET and EUR/RON                               | Solved,<br>Proposed | B3     |
| C1, C2 | US inflation and regimes; what is wrong in an AI answer?          | Proposed            | B1, B2 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

## Data used

| Series             | Source                   | Frequency           | Period    |
|--------------------|--------------------------|---------------------|-----------|
| Nile flow at Aswan | statsmodels (nile)       | yearly              | 1871–1970 |
| HICP, Romania      | Eurostat (prc_hicp_minr) | monthly, 2015 = 100 | 2005–2026 |
| CPI, United States | FRED (CPIAUCSL)          | monthly, SA         | 1947–2026 |
| S&P 500, BET       | EODHD                    | daily, close        | 2000–2026 |
| EUR/RON            | BNR reference rate       | daily               | 2005–2026 |

- Romanian inflation: monthly rate  $100\Delta \ln P_t$  minus its monthly means (seasonal adjustment), and the 12-month rate  $100(\ln P_t - \ln P_{t-12})$
- In the notebook: `gph`, `local_whittle`, `hurst_rs`, `arfima_exact_ml`, `monthly_rv`; no account or key is needed

## Prerequisites for Today (1/5): short and long memory

- $\rho(k)$ : the autocorrelation at lag  $k$  ([Chapter 1](#)); memory = how fast  $\rho(k)$  goes to zero
- **Short memory** (ARMA, [Chapter 2](#)):  $|\rho(k)| \leq Cr^k$ ,  $0 < r < 1$ ; the autocorrelations are summable
  - ▶  $C > 0$ : a constant;  $r$ : the decay rate per lag; AR(1):  $\rho(k) = \phi^k$
- **Long memory**:  $\rho(k) \sim Ck^{2d-1}$ ,  $0 < d < 1/2$ : hyperbolic decay,  $\sum_k \rho(k) = \infty$ 
  - ▶  $d$ : the memory parameter;  $\sim$ : the ratio of the two sides tends to 1 as  $k \rightarrow \infty$
  - ▶ equivalently, the spectral density has a pole at zero:  $f(\lambda) \sim G\lambda^{-2d}$  as  $\lambda \rightarrow 0$  ( $\lambda$ : frequency in radians;  $G > 0$ : a constant)
- **Hurst exponent**  $H = d + 1/2$ :  $H = 0.5$  no memory,  $H > 0.5$  persistence,  $H < 0.5$  anti-persistence
  - ▶ Hurst found  $H \approx 0.7$  for the Nile [Hurst \(1951\)](#)

## Prerequisites for Today (2/5): fractional differencing and ARFIMA

- Lag operator  $Lx_t = x_{t-1}$ ; the fractional difference  $(1 - L)^d$  for real  $d$  uses all past values
  - ▶  $(1 - L)^d = \sum_k \pi_k L^k$ , with  $\pi_0 = 1$ ,  $\pi_k = \pi_{k-1}(k - 1 - d)/k$ ;  $\pi_k$ : the weight of  $x_{t-k}$
  - ▶ the inverse  $(1 - L)^{-d} = \sum_k \psi_k L^k$ ,  $\psi_k = \psi_{k-1}(k - 1 + d)/k$ :  $\psi_k$  is the effect of a shock after  $k$  periods
- **ARFIMA**( $p, d, q$ ):  $\phi(L)(1 - L)^d(x_t - \mu) = \theta(L)\varepsilon_t$  [Granger and Joyeux \(1980\)](#), [Hosking \(1981\)](#)
  - ▶  $\phi(L)$ ,  $\theta(L)$ : the AR and MA polynomials ([Chapter 2](#));  $\mu$ : the mean;  $\varepsilon_t$ : white noise with variance  $\sigma^2$
- Ranges: invertible for  $d > -1/2$ ; stationary for  $d < 1/2$ ; mean-reverting for  $d < 1$ ;  $d = 1$  unit root
  - ▶ if  $1/2 \leq d < 3/2$ , difference once:  $(1 - L)x_t$  is ARFIMA with  $d - 1$

## Prerequisites for Today (3/5): ARFIMA(0, $d$ , 0) in numbers

- ▣ Autocorrelations by recursion:  $\rho(1) = d/(1 - d)$ ,  $\rho(k) = \rho(k - 1)(k - 1 + d)/(k - d)$
- ▣ Variance:  $\gamma(0) = \sigma^2 \Gamma(1 - 2d)/\Gamma(1 - d)^2$ 
  - ▶  $\Gamma(\cdot)$ : the gamma function,  $\Gamma(n) = (n - 1)!$  for integers; for  $d > 0$ ,  $\gamma(0) > \sigma^2$ : the memory accumulates past shocks
- ▣ One-step forecast:  $\hat{x}_{T+1} = \mu - \sum_{k \geq 1} \pi_k (x_{T+1-k} - \mu)$ 
  - ▶  $x_T$ : the last observation; since  $\pi_k < 0$  for  $k \geq 1$  when  $0 < d < 1$ , every past deviation from  $\mu$  pushes the forecast in its own direction

## Prerequisites for Today (4/5): estimators of $d$ and $H$

- ▣ **R/S**: in a block of  $n$  values,  $Y_k = \sum_{i \leq k} (x_i - \bar{x})$ ,  $R = \max Y_k - \min Y_k$ ,  $S$  the standard deviation (divisor  $n$ )
  - ▶  $\bar{x}$ : the block mean;  $H$  = slope of  $\log_{10}(R/S)_n$  on  $\log_{10} n$ ; **DFA**: slope of the detrended fluctuation  $\log F(n)$
- ▣ **GPH**: OLS of  $y_j = \log I(\lambda_j)$  on  $X_j = -\log(4 \sin^2(\lambda_j/2))$ ,  $j = 1, \dots, m$ ; slope =  $\hat{d}$ ,  $SE = \pi/\sqrt{24m}$ 
  - ▶  $I(\lambda_j)$ : the periodogram (a noisy estimate of  $f$ ) at the Fourier frequency  $\lambda_j = 2\pi j/T$ ;  $T$ : the number of observations;  $m$ : the number of low frequencies used
  - ▶ OLS slope =  $\sum(X_j - \bar{X})(y_j - \bar{y}) / \sum(X_j - \bar{X})^2$
- ▣ **Local Whittle** (LW) [Robinson \(1995\)](#): a local likelihood on the same  $m$  frequencies,  $SE = 1/(2\sqrt{m})$ ; bandwidth  $m = \lfloor T^{0.65} \rfloor$ 
  - ▶ **Exact ML** [Sowell \(1992\)](#): the full ARFIMA( $p, d, q$ ) likelihood; compare models by AIC and BIC

## Prerequisites for Today (5/5): volatility and spurious memory

- Returns  $r_t$  are almost uncorrelated;  $|r_t|$  and realised volatility have long memory [Ding, Granger and Engle \(1993\)](#)
  - ▶ monthly **realised volatility**:  $\sqrt{RV_m}$ ,  $RV_m = \sum_{t \in m} r_t^2$ , the sum of the squared daily returns of month  $m$ ; we model its logarithm
  - ▶ **shuffle test**: permute the days at random; real memory disappears, the distribution stays
- GARCH(1,1) as ARCH( $\infty$ ): weight  $\alpha\beta^{k-1}$  on  $\varepsilon_{t-k}^2$ ; FIGARCH [Baillie, Bollerslev and Mikkelsen \(1996\)](#): weights  $\approx k^{-1-d}$ 
  - ▶  $\varepsilon_{t-k}^2$ : the squared shock of  $k$  days ago;  $\alpha, \beta$ : the GARCH parameters ([Chapter 5](#));  $\sigma_t^2$ : today's conditional variance
  - ▶ with  $\phi = \beta = 0$ , FIGARCH weights are  $\lambda_k = -\pi_k$  of  $(1-L)^d$
- **Spurious long memory**: level shifts and rare regime switches give  $\hat{d} > 0$  for short-memory series [Diebold and Inoue \(2001\)](#); re-estimate after removing the breaks

## A1: fractional weights and the ACF [Solved]

### Task

- ▣ A series is ARFIMA(0, 0.25, 0) with  $\sigma^2 = 1$ .
- ▣ 1. Compute  $\pi_1, \dots, \pi_4$  of  $(1 - L)^{0.25}$ .
- ▣ 2. Compute  $\rho(1)$  and  $\rho(2)$ , and give  $H$  and the class of the process.
- ▣ 3. Compare  $\rho(5)$  with that of an AR(1) with the same  $\rho(1)$ .
- ▣ Report: four weights, two autocorrelations,  $H$ , two values of  $\rho(5)$ .

### Solution

- ▣ 1.  $\pi_1 = -0.25$ ;  $\pi_2 = -0.25 \cdot 0.75/2 = -0.0938$ ;  
 $\pi_3 = -0.0938 \cdot 1.75/3 = -0.0547$ ;  
 $\pi_4 = -0.0547 \cdot 2.75/4 = -0.0376$
- ▣ 2.  $\rho(1) = 0.25/0.75 = 0.333$ ;  $\rho(2) = 0.333 \cdot 1.25/1.75 = 0.238$ ;  
 $H = 0.75$ : stationary, long memory
- ▣ 3. ARFIMA:  $\rho(5) = 0.151$ ; AR(1):  $0.333^5 = 0.0041$ , about 37 times smaller

## A2: which range is $d$ in? [Proposed]

Solution discussed in class

### Task

- ▣ Two series: (a) ARFIMA(0, 0.6, 0); (b) ARFIMA(0, -0.3, 0). Model: A1.
- ▣ 1. Compute  $\pi_1$  and  $\pi_2$  of  $(1 - L)^d$  for each.
- ▣ 2. Say whether each series is stationary, invertible and mean-reverting.
- ▣ 3. For (b) compute  $\rho(1)$  and  $\rho(2)$ ; for (a) compute  $\rho(1)$  of the first difference.
- ▣ Report: four weights, two classifications, three autocorrelations.

A3:  $R/S$  by hand and  $H$  from three points [Solved]

## Task

- ▣ Block:  $x = (0.6, 0.4, 0.5, 0.3, -0.4, -0.6, -0.2, -0.5)$ . An  $R/S$  plot gives  $(R/S)_n = 4.1; 9.6; 22.4$  for  $n = 16, 64, 256$ .
- ▣ 1. Compute  $\bar{x}$ , the cumulative deviations  $Y_k$ ,  $R$ ,  $S$  and  $R/S$ .
- ▣ 2. Compute  $H$  as the slope of  $\log_{10}(R/S)_n$  on  $\log_{10} n$ , and  $d = H - 0.5$ .
- ▣ Report:  $R/S$ ,  $H$ ,  $d$  and one sentence.

## Solution

- ▣ 1.  $\bar{x} = 0.0125$ ;  $Y_k$ : 0.5875, 0.9750, 1.4625, 1.7500, 1.3375, 0.7250, 0.5125, 0;  $R = 1.750$ ,  $S = 0.457$ ,  $R/S = 3.83$
- ▣ the signs are grouped (four positive, then four negative): the range is large, as for a persistent series
- ▣ 2.  $\log_{10} n$ : 1.204, 1.806, 2.408;  $\log_{10}(R/S)$ : 0.613, 0.982, 1.350; slope  $H = 0.61$ ,  $d = 0.11$ : moderate persistence

## A4: a GPH slope by hand [Proposed]

Solution discussed in class

### Task

- ▣ For  $m = 4$  frequencies:  $X_j = 0.5; 1.6; 2.3; 3.0$  and  $\log l(\lambda_j) = 1.2; 1.8; 2.0; 2.5$ . Model: A3.
- ▣ 1. Compute  $\bar{X}$ ,  $\bar{y}$ ,  $\sum(X_j - \bar{X})(y_j - \bar{y})$  and  $\sum(X_j - \bar{X})^2$ .
- ▣ 2. Compute  $\hat{d}$  and its standard error  $\pi/\sqrt{24m}$ .
- ▣ 3. Test  $H_0: d = 0$  at 5% and comment on the size of  $m$ .
- ▣ Report: four sums,  $\hat{d}$ , its SE, a decision and one sentence.



## A5: a one-step ARFIMA forecast [Solved]

## Task

- ▣ ARFIMA(0, 0.4, 0) with  $\mu = 2$ ; the last four values:  $x_T = 3.0$ ,  $x_{T-1} = 2.5$ ,  $x_{T-2} = 2.8$ ,  $x_{T-3} = 1.5$ .
- ▣ 1. Write the first four weights  $-\pi_k$  of  $(1 - L)^{0.4}$ .
- ▣ 2. Compute  $\hat{x}_{T+1}$  using these four lags.
- ▣ 3. Compare with an AR(1) with the same  $\rho(1)$ .
- ▣ Report: four weights, two forecasts and one sentence.

## Solution

- ▣ 1.  $-\pi_k$ : 0.4; 0.12; 0.064; 0.0416
- ▣ 2.  

$$\hat{x}_{T+1} = 2 + 0.4(1.0) + 0.12(0.5) + 0.064(0.8) + 0.0416(-0.5) = 2 + 0.4000 + 0.0600 + 0.0512 - 0.0208 = 2.490$$
- ▣ 3. AR(1):  $\phi = \rho(1) = 0.4/0.6 = 0.667$ ,  
 $\hat{x}_{T+1} = 2 + 0.667 \cdot 1.0 = 2.667$ ; the AR(1) looks only at  $x_T$ , ARFIMA spreads the weight over the past (in practice over all  $T$  values)



## A6: how long does a volatility shock count? [Proposed]

Solution discussed in class

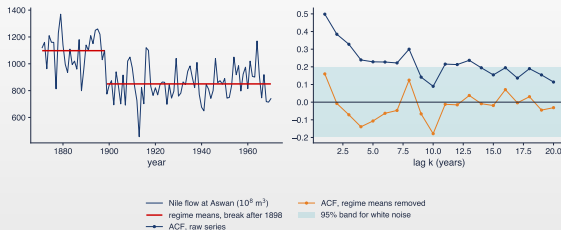
### Task

- ▣ GARCH(1,1) with  $\alpha = 0.10$ ,  $\beta = 0.88$ ; FIGARCH(0,  $d$ , 0) with  $d = 0.4$  (weights  $\lambda_k = -\pi_k$ ). Model: A5.
- ▣ 1. Compute the weight of  $\varepsilon_{t-k}^2$  in  $\sigma_t^2$  for  $k = 1, 2, 3$  in both models.
- ▣ 2. Compute both weights at  $k = 20$  and  $k = 100$ ; for FIGARCH use  $\lambda_k \approx d k^{-1-d} / \Gamma(1-d)$ ,  $\Gamma(0.6) = 1.489$ .
- ▣ 3. Explain which model keeps a crisis in today's volatility longer.
- ▣ Report: ten weights and one sentence.

## B1: does the Nile have long memory? [Solved]

- **Question:** is the persistence of the Nile flow long memory, or the effect of the change in 1898?
- **Inputs:** yearly flow at Aswan, 1871–1970 ( $T = 100$ )
- **Tasks**
  1. Plot the series and its ACF up to lag 20.
  2. Estimate  $H$  by R/S,  $d$  by GPH and local Whittle ( $m = \lfloor T^{0.65} \rfloor$ ) and by exact ML of ARFIMA(0,  $d$ , 0).
  3. Subtract the mean of 1871–1898 and the mean of 1899–1970 and repeat step 2.
  4. Interpretation: does the Nile have long memory?
- **Report:** two ACF values, four estimates before and after, one sentence
- **Notebook:** section B1

## B1: solution [Solved]



- Raw:  $\hat{\rho}(1) = 0.50$ ,  $\hat{\rho}(10) = 0.09$ ;  $H_{R/S} = 0.78$ ; GPH 0.44 (SE 0.15), LW 0.40 (SE 0.11), ML 0.36 (SE 0.07);  $m = 19$
- Means 1098 and 850; after removing them:  $\hat{\rho}(1) = 0.16$ ;  $H_{R/S} = 0.63$ ; GPH  $-0.07$ , LW  $-0.17$ , ML 0.05 (SE 0.10)
- Interpretation: for 1871–1970 the evidence of long memory comes from one break;  $R/S$  stays above 0.5 because it is biased upwards in short samples

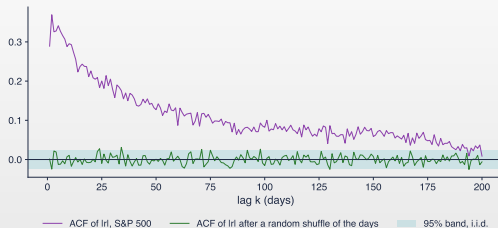
## B2: long memory in Romanian inflation [Proposed]

- **Question:** how persistent is Romanian inflation, and does the way we measure inflation change the answer?
- **Inputs:** monthly HICP, 2005–2026: the monthly rate seasonally adjusted with the monthly means, and the 12-month rate; model: B1
- **Tasks**
  1. Estimate  $d$  of the monthly rate by GPH and local Whittle with  $m = \lfloor T^a \rfloor$ ,  $a = 0.5; 0.65; 0.8$ .
  2. Fit ARFIMA(0,  $d$ , 0) and ARMA(1,1) by exact ML and compare their BIC.
  3. Estimate  $d$  of the 12-month rate by local Whittle and compare the two ACFs.
  4. Interpretation: why does the 12-month rate give  $d$  close to 1?
- **Report:** six estimates, two BIC values, one  $d$ , two sentences
- **Notebook:** section B2

## B3: memory in S&P 500 volatility [Solved]

- **Question:** where is the memory of stock returns: in the returns or in their size?
- **Inputs:** daily log returns of the S&P 500 since 2000 and their monthly realised volatility
- **Tasks**
  1. Estimate  $d$  of  $r_t$  and of  $|r_t|$  by local Whittle and GPH.
  2. Shuffle the days at random and re-estimate  $d$  of  $|r_t|$ ; plot both ACFs of  $|r_t|$ .
  3. Estimate  $d$  of the monthly log realised volatility.
  4. Interpretation: what does the shuffle test show?
- **Report:** eight estimates, the chart and one sentence
- **Notebook:** section B3

## B3: solution [Solved]



- $T = 6714$  days;  $r_t$ : LW  $-0.01$ , GPH  $0.02$  (SE  $0.03$ ,  $0.04$ );  $|r_t|$ : LW  $0.53$ , GPH  $0.56$
- Shuffled  $|r_t|$ : LW  $-0.03$ , GPH  $-0.03$ ; ACF of  $|r_t|$ :  $0.289$  (lag 1),  $0.127$  (lag 50),  $0.009$  (lag 200); log RV (321 months): LW  $0.41$ , GPH  $0.37$
- Interpretation: the shuffle keeps every daily value and destroys their order; the memory disappears, so it lives in the clustering of calm and turbulent periods, not in the distribution of returns

## B4: BET and EUR/RON [Proposed]

- **Question:** do the BET and the EUR/RON rate show the same memory, and is it stable before and after the 2008 crisis?
- **Inputs:** daily log returns of the BET (since 2000) and of the EUR/RON reference rate (since July 2005); model: B3
- **Tasks**
  1. Estimate  $d$  of  $r_t$  and  $|r_t|$  by local Whittle on the full sample.
  2. Repeat on the subsamples up to 2007 and 2010–2026.
  3. Compare the estimates for  $r_t$  with the 95% Monte Carlo band of i.i.d. series of 1900 days.
  4. Interpretation: is the full-sample memory of BET returns real?
- **Report:** twelve estimates, one band and two sentences
- **Notebook:** section B4

## C1: US inflation and monetary regimes [Proposed]

- ▣ **Question:** is the long memory of US inflation a property of inflation, or of the changes in monetary policy?
- ▣ **Inputs:** US monthly CPI inflation, annualised, 1947–2026; regimes 1947–1984, 1985–2019, 2020–2026; models: B1, B2
- ▣ **Tasks**
  1. Estimate  $d$  by local Whittle on the full sample and in each regime.
  2. Remove the three regime means and re-estimate  $d$ .
  3. Propose one more check (a break test, a Markov-switching model, a forecast comparison).
  4. Interpretation: can a constant  $d$  describe 80 years of inflation?
- ▣ **Report:** five estimates, one check and a project plan
- ▣ **Notebook:** section C1

## C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about the long memory of Romanian inflation and of the S&P 500. The answer:
- ▣ (a) Monthly inflation has  $d = 0.31$ , so its Hurst exponent is  $H = d - 0.5 < 0$ : it is anti-persistent.
- ▣ (b) The 12-month rate gives  $d = 1.00$ , so Romanian inflation has a unit root.
- ▣ (c) With  $d = 0.31$  the series is stationary and its shocks die out.
- ▣ (d) Use `statsmodels.tsa.arima.model.ARIMA(x, order=(0, 0.31, 0))` to fit the ARFIMA model.
- ▣ (e) S&P 500 returns have  $d = -0.01$  and  $|r|$  has  $d = 0.53$ , so returns are predictable.
- ▣ (f) A significant GPH estimate proves long memory.
- ▣ **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- ▣ **Report:** a list of six verdicts with one line of justification each

## Key takeaways

- ▣  $(1 - L)^d$ : weights by the recursion  $\pi_k = \pi_{k-1}(k - 1 - d)/k$ ; ARFIMA(0,  $d$ , 0):  $\rho(1) = d/(1 - d)$
- ▣  $H = d + 1/2$ ; R/S and GPH are slopes of log-log regressions
- ▣ Romanian monthly inflation:  $d \approx 0.3$ ; S&P 500: memory in  $|r_t|$ , not in  $r_t$
- ▣ Breaks can create long memory: the Nile, BET returns, US inflation
- ▣ An AI answer is a draft: check  $H$  against  $d$ , the library, and the data transformation

## After the seminar

- Lecture 8 develops each topic of today
  - ▶ hyperbolic decay, fractional differencing, ARFIMA, the estimators, forecasting, volatility, spurious memory
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation memory and monetary regimes in Romania
- Reading: [Baillie \(1996\)](#); [Graves et al. \(2017\)](#); [Ding, Granger and Engle \(1993\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

## References

- ▣ Baillie, R. T. (1996). Long memory processes and fractional integration in econometrics. *Journal of Econometrics*, 73(1), 5–59. doi:10.1016/0304-4076(95)01732-1
- ▣ Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30. doi:10.1016/S0304-4076(95)01749-6
- ▣ Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- ▣ Ding, Z., Granger, C. W. J., & Engle, R. F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106. doi:10.1016/0927-5398(93)90006-D
- ▣ Granger, C. W. J., & Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29. doi:10.1111/j.1467-9892.1980.tb00297.x
- ▣ Graves, T., Gramacy, R., Watkins, N., & Franzke, C. (2017). A brief history of long memory: Hurst, Mandelbrot and the road to ARFIMA, 1951–1980. *Entropy*, 19(9), 437. doi:10.3390/e19090437
- ▣ Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165
- ▣ Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799. doi:10.1061/TACEAT.0006518
- ▣ Robinson, P. M. (1995). Gaussian semiparametric estimation of long range dependence. *The Annals of Statistics*, 23(5), 1630–1661. doi:10.1214/aos/1176324317
- ▣ Sowell, F. (1992). Maximum likelihood estimation of stationary univariate fractionally integrated time series models. *Journal of Econometrics*, 53(1–3), 165–188. doi:10.1016/0304-4076(92)90084-5